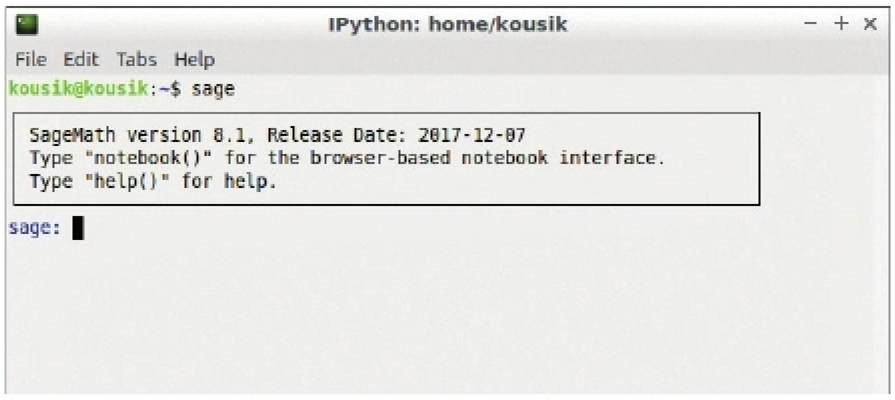

Figure 1: SageMath CLI

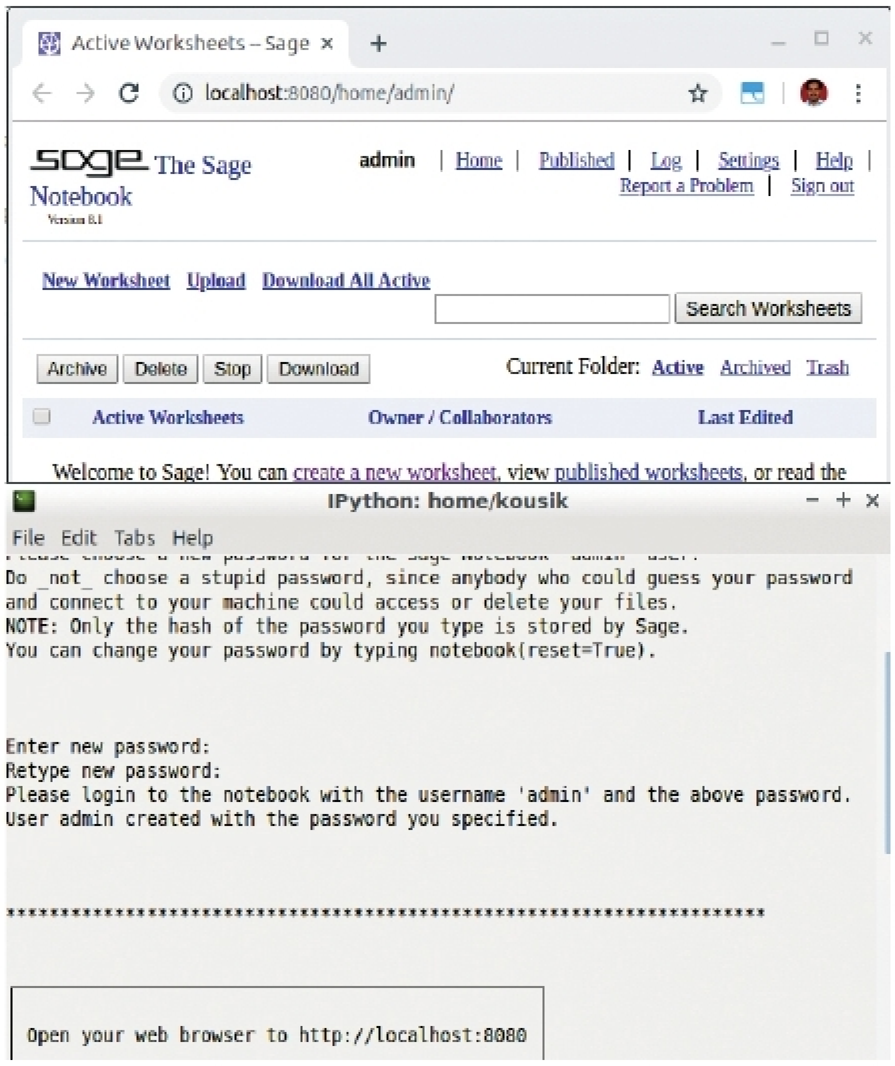

Figure 2: *notebook()* command output

The basic integer operations are:

```
integer division a // b
remainder a % b
quotient and remainder divmod(a,b)
factorial n! factorial(n)
binomial coefficient n k binomial(n,k)
```

The usual functions on real numbers and complex numbers are:

```
integer part floor(a)
absolute value, modulus abs(a)
elementary functions sin, cos
```

2D or 3D graphs can also be plotted by Sage. To plot a 3D graph, you need to have Java configured in your system. The plot of a 2D graph is shown in Figure 4.

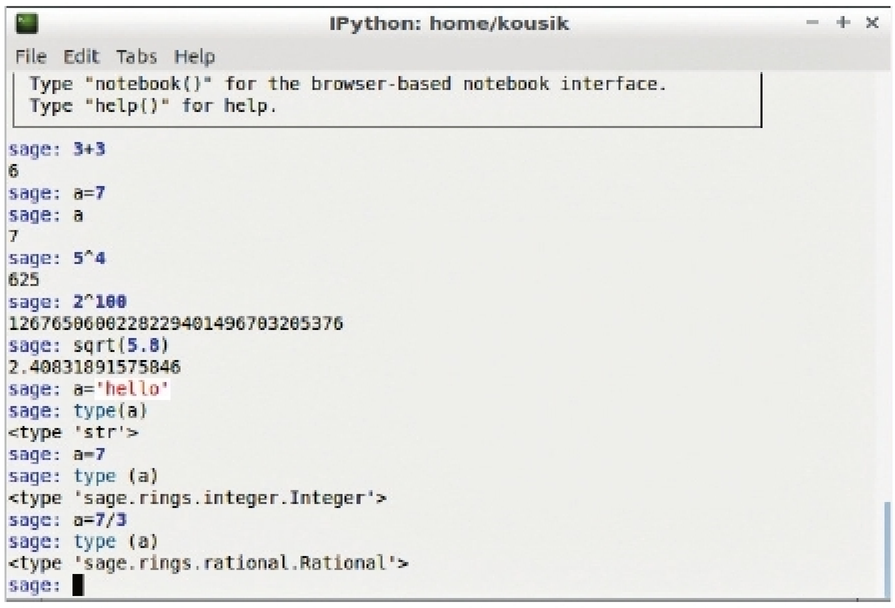

Figure 3: Basic operations

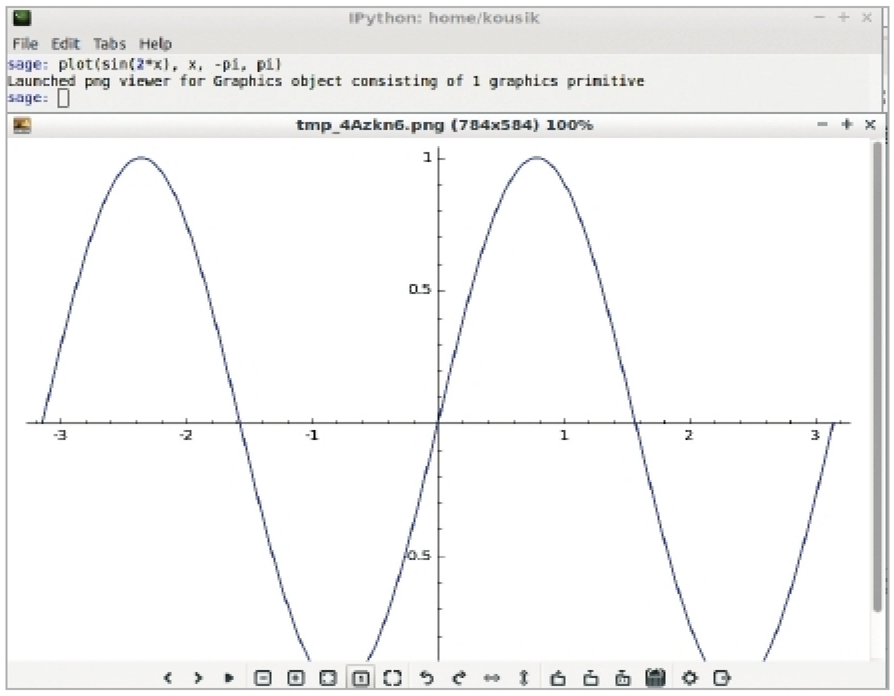

Figure 4: Plotting a graph with SageMath

To get help, you have to type the following command:

```
sage: keyword?
Here I run
sage: cot?
```

You can get more information from *http://doc.sagemath.org/pdf/en/tutorial/SageTutorial.pdf.*

You can learn more about Sage usage from *http://doc.sagemath.org/pdf/en/tutorial/SageTutorial.pdf.* A complete free book on SageMath is available at *http://sagebook.gforge.inria.fr/english.html.* END

References

[1] *http://www.sagemath.org/*
[2] *https://en.wikipedia.org/wiki/SageMath*

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